SESSION 2 — MORNING

Microeconomics

How firms, markets and prices work

Learning Objectives

Supply & Demand

Explain market equilibrium and predict shifts from changes in price, income and other factors.

Elasticity

Compute and interpret price elasticity of demand and supply. Apply to pricing strategy.

Market Structures

Distinguish perfect competition, monopolistic competition, oligopoly and monopoly with their profit implications.

Costs & Profit Maximisation

Understand fixed vs variable costs, marginal analysis, breakeven and shutdown decisions.

Supply & Demand — Market Equilibrium

Markets coordinate millions of decisions through one mechanism: price. Price moves until quantity supplied = quantity demanded.

D↓

Law of Demand

As price rises, quantity demanded falls (ceteris paribus). Exceptions: Giffen goods, Veblen (luxury) goods.

S↑

Law of Supply

As price rises, quantity supplied rises. Higher price makes production more profitable — more suppliers enter.

P*

Equilibrium

Market clears where S = D. Surplus: price is above P* (producers must cut). Shortage: price is below P* (consumers bid up).

Demand Shifters (non-price factors that move the CURVE)
Income • Prices of substitutes & complements • Consumer tastes & preferences • Expectations about future prices • Number of buyers in the market

Price Elasticity of Demand (PED)

$$PED = \%\,\Delta \text{ Quantity Demanded} \;/\; \%\,\Delta \text{ Price} = (\Delta Q/Q) \;/\; (\Delta P/P)$$

PED is always negative (demand slopes down) — we usually quote the absolute value.

Value	Type	Meaning	Examples	Revenue Impact
|PED| > 1	Elastic	Consumers very price-sensitive	Luxury goods, many substitutes	Cut price → revenue rises
|PED| = 1	Unit elastic	% changes are equal	Theoretical benchmark	Price change → revenue unchanged
|PED| < 1	Inelastic	Consumers less price-sensitive	Necessities, few substitutes	Raise price → revenue rises
|PED| = 0	Perfectly inelastic	Quantity fixed regardless of price	Life-saving drugs (short-run)	Any price → same quantity

WORKED EXAMPLE

Price of coffee rises from £3.00 to £3.30 (+10%). Quantity sold falls from 200 to 186 cups (−7%).

PED = −7% / +10% = −0.70 → |PED| = 0.70 < 1 → Inelastic. Revenue rises when price rises.

Revenue before: 200 × £3.00 = £600. Revenue after: 186 × £3.30 = £613.80 ✓

Cross-Price & Income Elasticity

Cross-Price Elasticity (CPE)

% Δ Quantity of A / % Δ Price of B

▶ CPE > 0: Substitutes — higher price of B raises demand for A (tea vs coffee)

▶ CPE < 0: Complements — higher price of B reduces demand for A (petrol & cars)

▶ CPE = 0: Unrelated goods

Finance use: A company's pricing power depends on the availability of substitutes. High CPE with substitutes = low pricing power = compressed margins.

Income Elasticity (YED)

% Δ Quantity Demanded / % Δ Income

▶ YED > 1: Luxury goods — demand rises faster than income (sports cars, fine dining)

▶ 0 < YED < 1: Normal goods — demand rises with income (clothing, electronics)

▶ YED < 0: Inferior goods — demand falls as income rises (budget supermarket food)

Finance use: Luxury goods companies (LVMH, Richemont) benefit from rising global wealth. Inferior goods businesses are recession-resistant.

SECTION 02

Costs & Market Structures

Cost Structures — The Building Blocks

Understanding costs is essential for margin analysis, breakeven modelling and evaluating competitive moats.

Fixed Costs (FC)

Unchanged with output. Rent, depreciation, executive salaries. Creates operating leverage — FC/VC mix determines profit volatility.

Variable Costs (VC)

Rise directly with output. Raw materials, direct labour, packaging. Variable cost ratio = VC/Revenue = 1 − Gross Margin.

Marginal Cost (MC)

Cost of one additional unit. MC = ΔTC/ΔQ.
Profit max rule: produce until MR = MC. If MC rises, set output where MC = Price.

Average Total Cost (ATC)

TC/Q. The U-shaped ATC curve shows economies then diseconomies of scale.
Minimum ATC = efficient scale. Below it: growing is beneficial.

BREAKEVEN

Breakeven Q = Fixed Costs / (Price − Variable Cost per unit) = FC / Contribution Margin
Ex: FC = £100,000, Price = £50, VC/unit = £30 → Contribution = £20 → BEP = 100,000/20 = 5,000 units

Market Structures — Implications for Profitability

The competitive environment shapes a company's long-run earnings potential — core to equity analysis.

Structure	# Firms	Product	Pricing	LR Profit	Examples	An No
Perfect Competition	Many	Identical	Price taker P=MC	Zero	Wheat, FX markets	⍰ mo
Monopolistic Competition	Many	Differentiated	Some power P>MC	Zero LR	Restaurants, apparel	So br va
Oligopoly	Few	Similar/diff	Interdependent	Positive (barriers)	Airlines, telecoms	Co ior
Monopoly	One	Unique	Price maker	Positive	Natural utilities	Re ris

KEY INSIGHT — Why Market Structure Matters for Valuation

Higher barriers to entry = more durable profits = justified premium P/E multiple.
Oligopoly returns are often 2-3× higher than perfect competition over the long run.
Analysts use Porter's Five Forces to map structure to profitability drivers.

Porter's Five Forces — Competitive Intensity

Michael Porter's framework translates market structure theory into a practical industry analysis toolkit.

1. Competitive Rivalry

Intensity of existing competition. Price wars, product innovation, advertising battles. HIGH rivalry → compressed margins.

Industry with 5 airlines vs one with 2 — very different pricing power.

2. Threat of New Entrants

Ease of entering the market. Low barriers → profit erosion over time. HIGH barriers → sustainable ROE.

Capital requirements, regulatory licences, brand loyalty, switching costs.

3. Buyer Bargaining Power

Powerful buyers push prices down. Concentrated customer base, easy switching, low differentiation amplify power.

Supermarkets vs FMCG suppliers. Automakers vs steel companies.

4. Supplier Bargaining Power

Strong suppliers raise input costs. Few alternatives, critical inputs, forward integration threats.

OPEC vs refiners. Semiconductor fab vs consumer electronics.

5. Threat of Substitutes

Alternative products cap pricing. Technology creates new substitutes. High substitute threat → ceiling on prices.

Streaming vs cinema. EVs vs petrol cars. Solar vs utility grids.

Consumer & Producer Surplus — Who Gains from Trade?

Every voluntary trade creates value. Surplus measures how much.

Consumer Surplus

Willingness to pay (WTP) MINUS actual price paid.

Area above market price, below demand curve.

Ex: You value a concert ticket at £150. You pay £80. CS = £70.

Higher price → CS falls. Price controls (ceilings) aim to protect CS at cost of reducing PS and total surplus.

Producer Surplus

Price received MINUS minimum acceptable price (MC).

Area below market price, above supply curve.

Ex: You sell a phone for £500. Minimum you'd accept = £350. PS = £150.

Lower price → PS falls. Taxes create deadweight loss by reducing both CS and PS.

Deadweight Loss = lost surplus from trades that don't happen due to price distortions, taxes, or market power.

PRACTICE PROBLEMS

Try before turning the page.

Q1	iPhone price rises from £999 to £1,099. Sales fall from 200,000 to 188,000 units. Calculate PED and comment on revenue impact.
Q2	A firm has FC=£80,000, VC/unit=£25, selling price=£45. What is the breakeven quantity? What profit at 6,000 units?
Q3	An analyst says 'this is a market with high barriers to entry and inelastic demand.' What does this suggest about margins?
Q4	Coffee and cream have CPE = −0.8. If cream prices rise 20%, what happens to coffee demand? Are they substitutes or complements?
Q5	Using Porter, assess: online streaming platform, existing players (Netflix, Amazon, Disney+), minimal switching cost for consumers.

Practice — Answers

A1 PED = (−12,000/200,000) / (100/999) = −6% / +10% = −0.60. Inelastic. Revenue: before=£199.8m, after=£206.6m. Revenue RISES.

A2 BEP = 80,000/(45−25) = 80,000/20 = 4,000 units. Profit at 6,000 = (6,000×20) − 80,000 = 120,000−80,000 = £40,000.

A3 High barriers protect long-run profits (no erosion from new entrants). Inelastic demand means pricing power. Expect high, durable margins → premium valuation multiple justified.

A4 CPE = −0.8 → complements. Demand change = −0.8 × 20% = −16%. Coffee demand falls 16%. They are complements (negative CPE).

A5 High competitive rivalry (Netflix, Amazon, Disney+). Low buyer switching costs = high buyer power. Threat from new entrants (Apple, Paramount). Strong substitute threat (gaming, social media). Porter suggests WEAK competitive position → price pressure → margin compression.

KEY TAKEAWAYS

Session 3 — Microeconomics

- Markets clear at P* where S = D — price signals coordinate decisions
- PED < 1: inelastic → raise price to grow revenue (utilities, drugs)
- Market structure: oligopoly/monopoly → durable profits, justified premium PE
- Profit max: produce where MR = MC — the universal rule
- Porter's 5 Forces maps competitive intensity to long-run profit potential
- Consumer + Producer Surplus = total welfare from trade